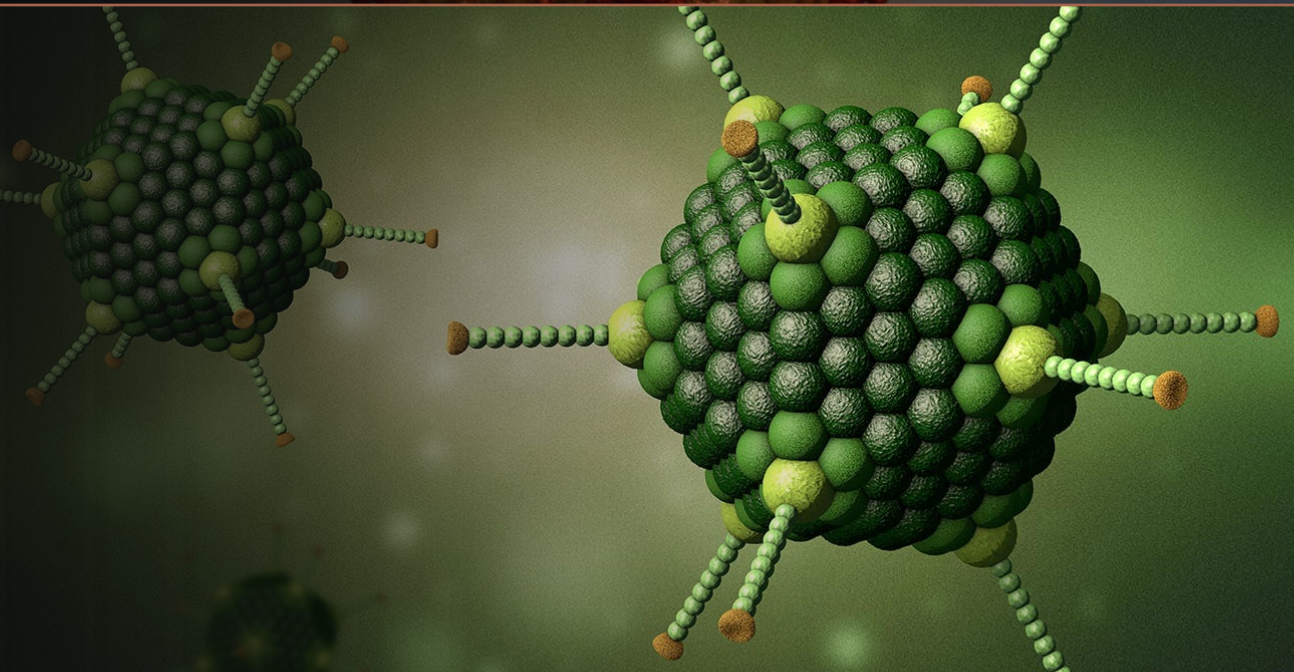
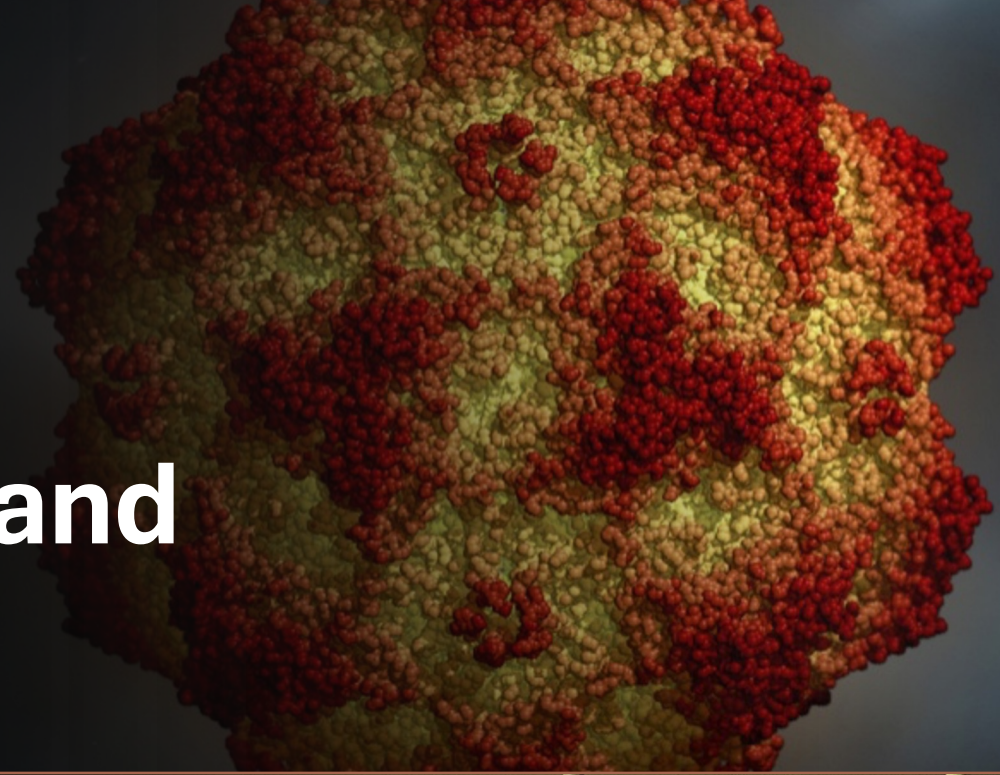


Family *Parvoviridae* and Family *Adenoviridae*

Wilasinee Surasa, DVM, M.Sc.

Lecturer, Faculty of Veterinary Sciences

Maha Sarakham University



วัตถุประสงค์

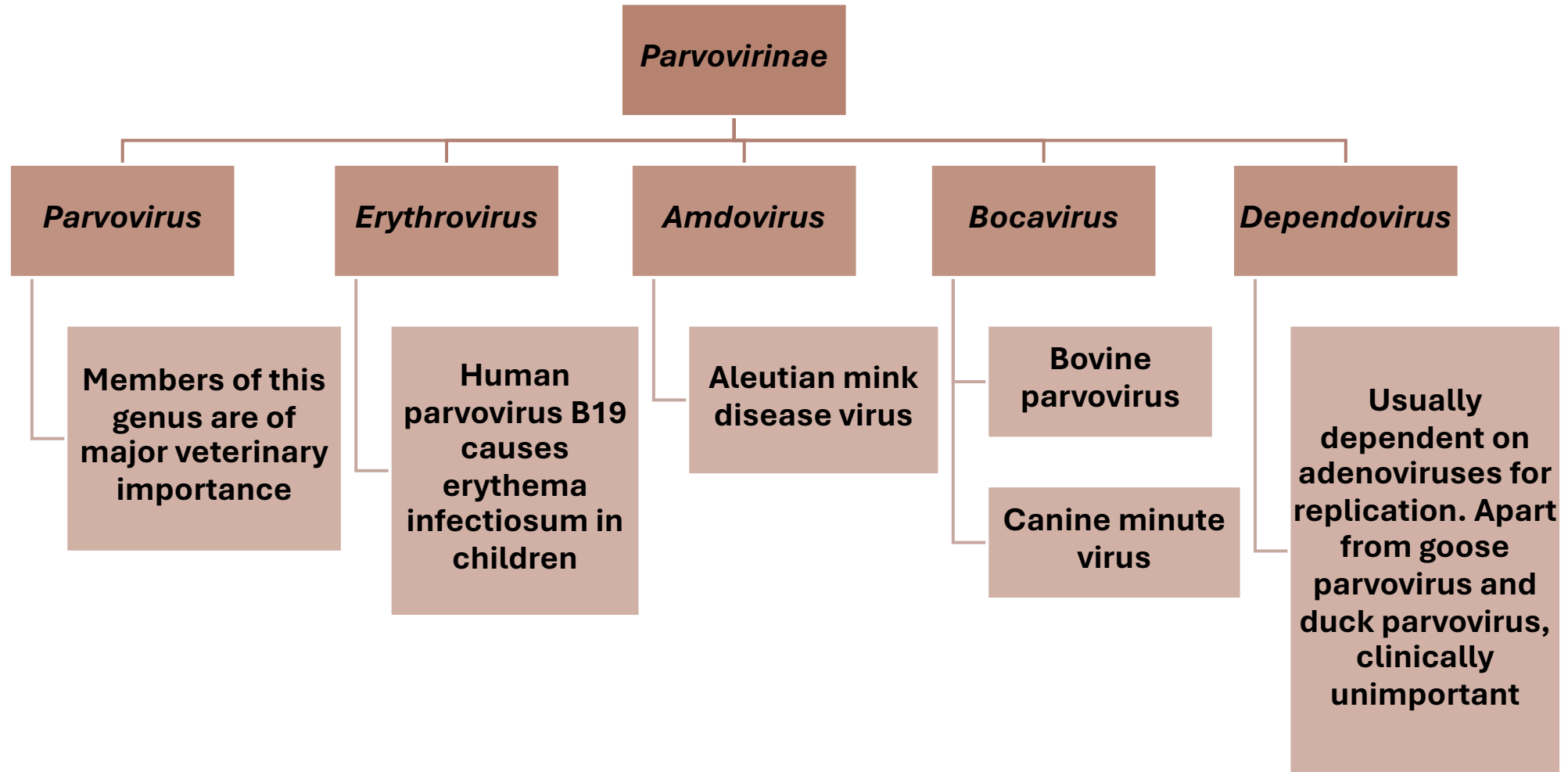
- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Family *Parvoviridae* และ Family *Adenoviridae* ดังนี้
 - ลักษณะและชีววิทยาของไวรัส
 - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
 - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
 - การควบคุมและการป้องกันโรค



Family *Parvoviridae*

- Taxonomic structure of the family
- Family *Parvoviridae*
 - Subfamily *Parvovirinae* :
viruses of vertebrates
 - Genus *Parvovirus*
 - Genus *Erythrovirus*
 - Genus *Dependovirus*
 - Genus *Amdovirus*
 - Genus *Bocavirus*
 - Subfamily *Densovirinae* :
viruses of insects

The genera of *Parvovirinae*



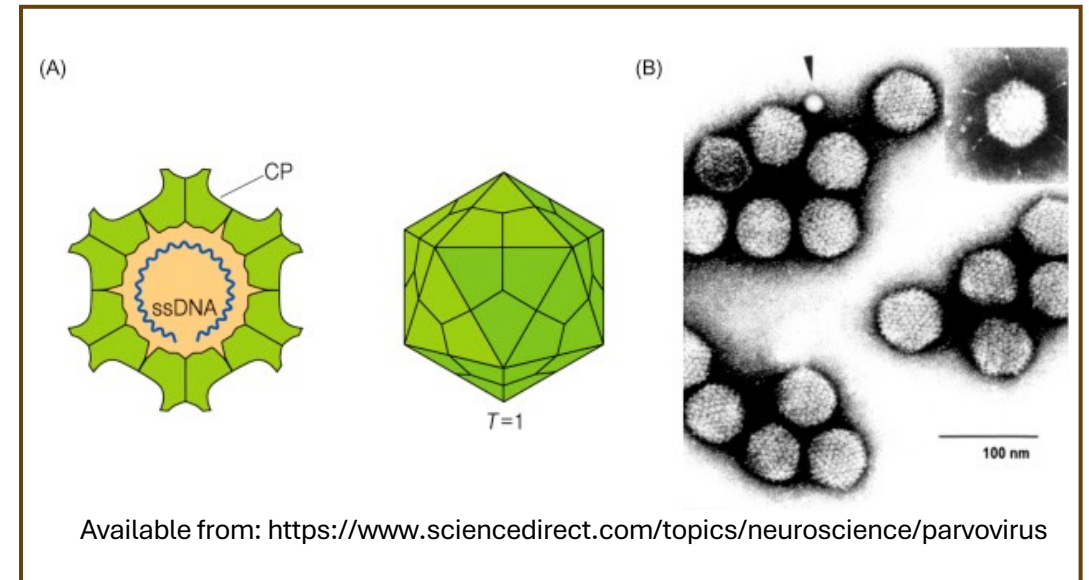
Manifestations of Parvovirus diseases in animals (MacLachlan and Dubovi, 2011,p.226)

virus	disease
Feline panleukopenia virus	Generalized disease in kittens, with panleukopenia, enteritis; cerebellar hypoplasia
Canine parvovirus 1 (minute virus of canines)	Minimal
Canine parvovirus 2 (subtypes 2a, 2b, 2c)	Generalized disease in puppies; enteritis, myocarditis (rarely), lymphopenia
Porcine parvovirus	Stillbirth, abortion, fetal death, mummification, infertility
Mink enteritis virus	Leukopenia, enteritis
Aleutian mink disease virus	Chronic immune complex disease, encephalopathy. Interstitial pneumonia in neonates
Mouse parvoviruses, minute virus of mice, rat parvoviruses, H-1 virus of rats	Subclinical or persistent infection; congenital fetal malformation; hemorrhagic syndrome in rats
Goose parvovirus	Hepatitis, myocarditis, myositis
Duck parvovirus	Hepatitis, myocarditis, myositis

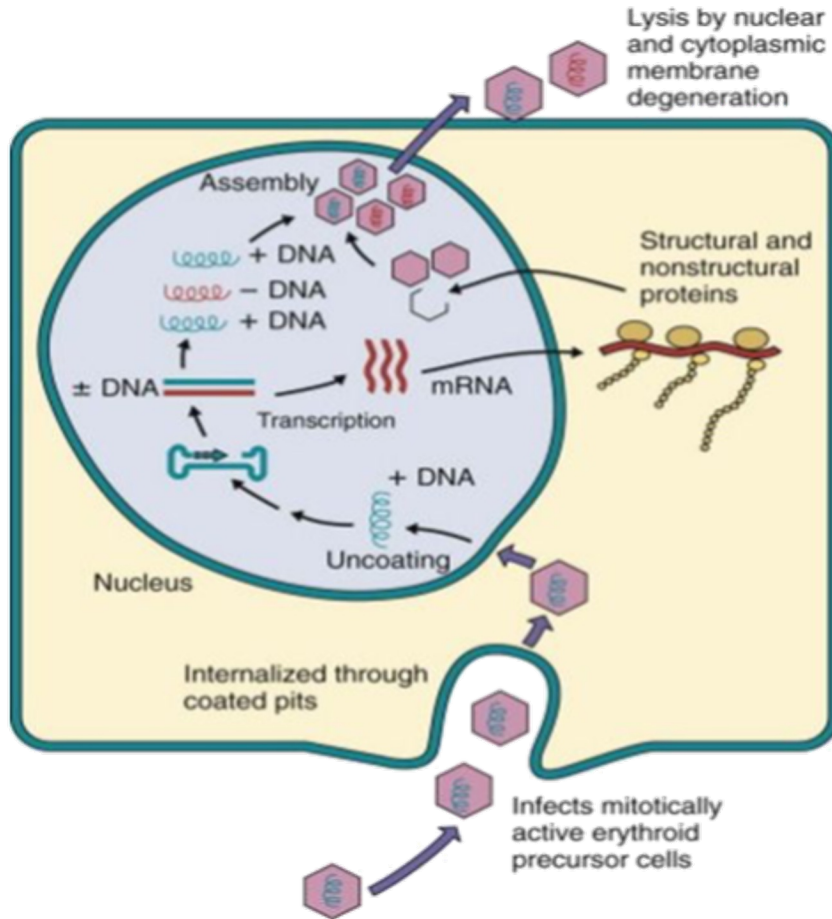
Virion properties



- Small, 18-26 nm diameter
- Icosahedral symmetry and non-enveloped
- single-stranded DNA viruses – linear genome
- Replicate in nucleus, form intranuclear inclusion bodies (ติดสี eosinophilic)
- Stable in the environment
- Resistant to heat, lipid solvents, disinfectants and pH changes
 - Heat: 56°C for more than 60 minutes
 - pH: 3–9
- Many have hemagglutinating activity



Viral replication



- Replicate only in the nuclei of dividing host cells
- **Baltimore classification**
 - Class II virus: single stranded DNA; ssDNA (+/-)
- After entering a cell → nucleus
 - Virion uncoated
 - Single-stranded DNA (+) → double stranded DNA (+/-)
 - By cellular DNA polymerases
 - Transcription → mRNA → structure and non-structure protein
 - Progeny ssDNA+protein → new virion
 - Cell lysis by nuclear and cytoplasmic membrane degeneration → virions released

Clinical infections

- Infect many domestic and wild animals
- Pathogenesis from viral replication
 - acute systemic disease
- Most severe in fetuses (after transplacental infection) and neonate
 - Cerebellar hypoplasia (fetus or neonate)
 - Myocarditis (ลูกสุนัข 3-8 weeks, ลูกห่าน 0-2 weeks)
 - Hepatitis (ลูกห่าน)



The most important parvoviral diseases of domestic mammals

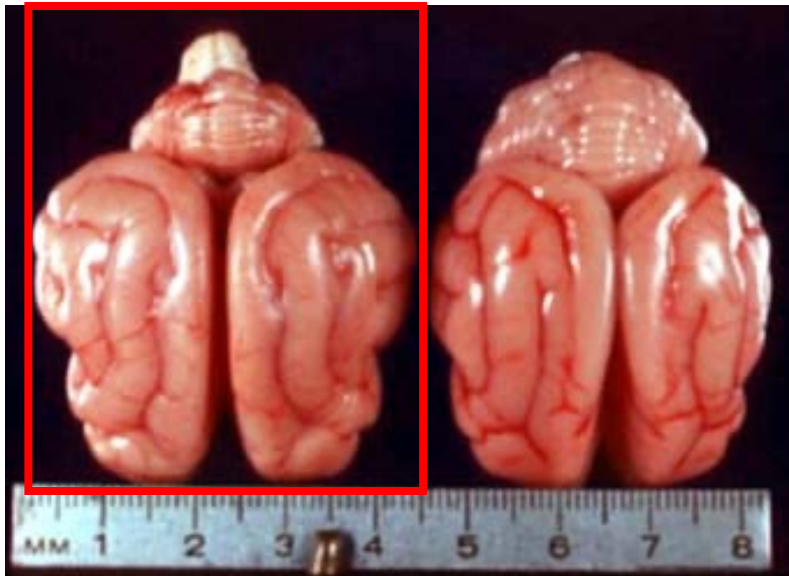
- Feline panleukopenia
- Canine parvovirus infection
- Porcine parvovirus infection



Feline panleukopenia virus

- Feline infectious enteritis or Feline distemper
- Highly contagious disease
 - Direct contact with infected cats or via fomites (bedding, food dishes)
 - Mechanical vector; fleas and human
- Aero-nasal and oral transmission
- Virus is shed in feces, vomit, urine and saliva
- Most common in kittens (หลังหย่านม ตอณภูมิคุ้มกันจากแม่ลดลง)
 - But cat of all ages are susceptible
- Incubation period: ~ 5 days (rang 2-10 days)





Clinical signs

- Fever, lassitude, inappetence, a rough coat, bloody diarrhea
 - Perinatal or in-utero infection
 - Kitten can cause abnormal development of cerebellum (cerebellar hypoplasia/atrophy syndrome)
 - Ataxic when kittens become ambulatory around 3 weeks of age (so called spastic or wobbly cat syndrome)

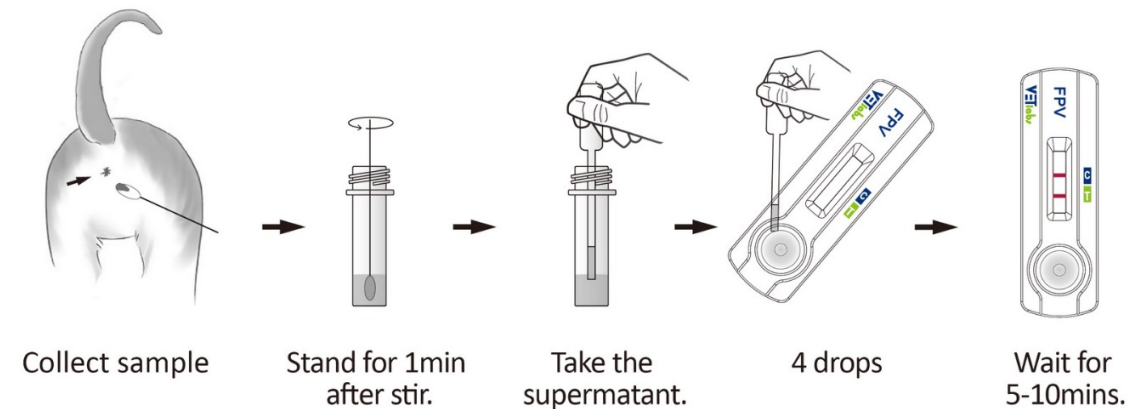
Pathogenesis

- Pharyngeal lymphoid tissue > blood circulation > WBC (lymphocyte, granulocyte, monocyte), platelets in blood circulation and lymphoid organ-Payer's patches > **panleukopenia**
- **Gross and histopathology**
 - สูญเสีย villi tip, villi สั้นดูดำซึมไม่ได้ > bloody diarrhea > dehydration (villi สั้นที่อและอาจมีเซลล์ตายที่ปลายยอด villi, อาจพบ N/IC ในเซลล์ใกล้เคียงของ crypt)

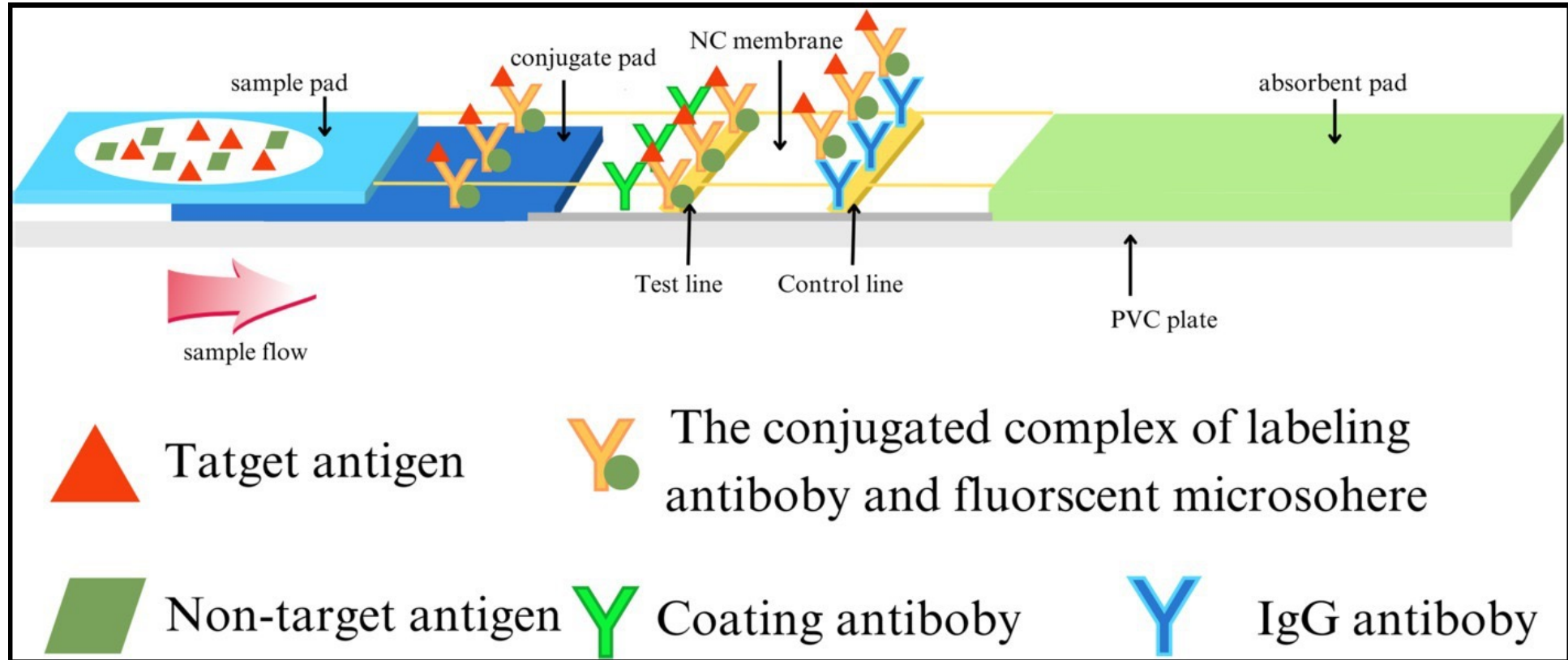


Diagnosis

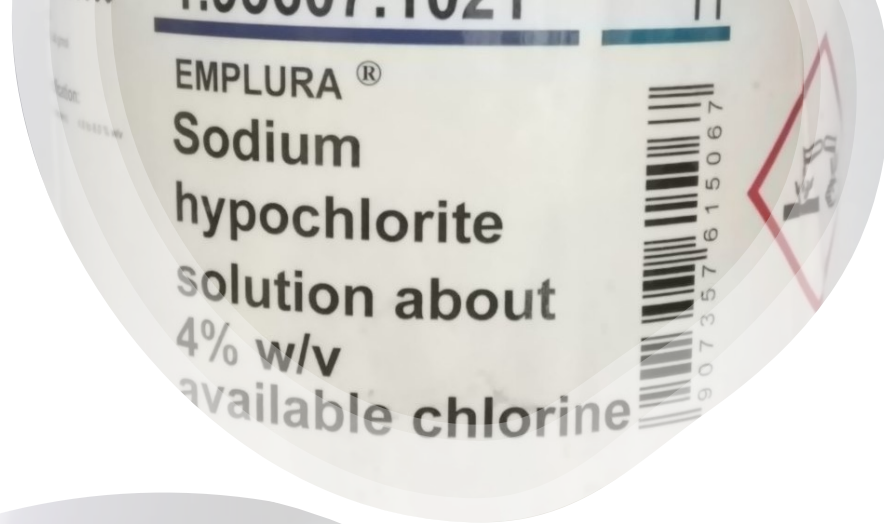
- Clinical signs, hematological data, and postmortem findings
 - Characteristic and sufficient for presumptive diagnosis
- The usual confirmatory tests
 - PCR assay > Detect viral DNA in feces or tissues
 - Ag-capture enzyme immunoassay or Immunofluorescence > Detect Ag in tissues
- Virus isolation or Hemagglutination
- Serologic diagnosis
 - Hemagglutination-inhibition assays
 - Enzyme immunoassay
 - Indirect immunofluorescence



Immunochromatographic assay



<https://www.frontiersin.org/journals/veterinary-science/articles/10.3389/fvets.2024.1413420/full>



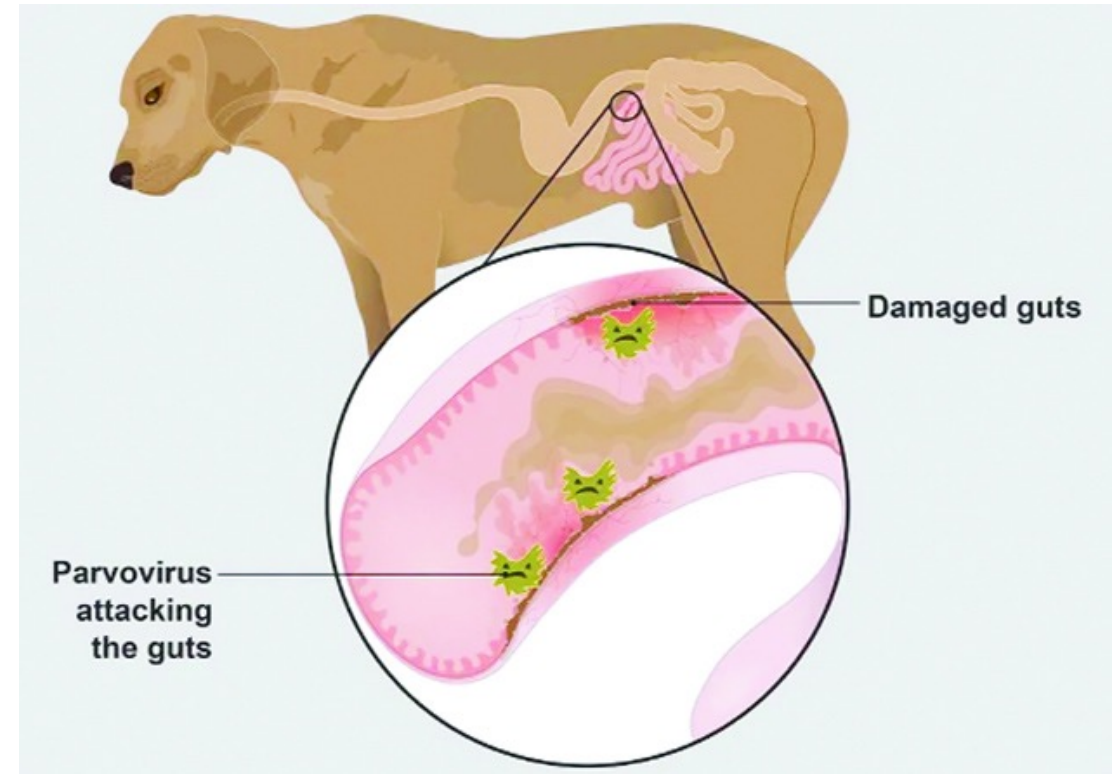
Control and prevention

- Vaccination***
 - Live vaccines are safe and potentially better than inactivated vaccine
- Isolate incoming cat for 2 weeks
 - Sick cat should remove and isolate
 - Vaccine should be used rigorously
- Disinfectant; 1% sodium hypochlorite



Canine parvovirus type 2

- With 3 major variant (2a, 2b, 2c)
- Fecal-oral route transmission
- Highly contagious and very stable in environment
- All members of the family Canidae (dogs, wolves, foxes, coyote)
 - 6 weeks - 6 months of age
- Very important cause of infectious diarrhea in both wild and domestic canids
- CPV replicates in several lymphoid tissues and intestinal epithelium of dog





Clinical signs

- Generalize neonatal death
- Panleukopenia enteritis (คล้ายในแมว)
- Myocarditis (มักพบในลูกสุนัขอายุ 1 สัปดาห์)
 - Acute HF and sudden death / survive puppy > developed cardiomyopathy at 4-8 weeks of age
- Anorexia, lethargy, vomit, bloody diarrhea and fetid smell, death > electrolyte imbalance, severe dehydration
- Feces > streaked with blood or and frankly hemorrhagic



Pathogenesis

- Pathogenesis similar to Feline panleukopenia virus
 - Absence of cerebellar hypoplasia
 - Occurrence of myocarditis in puppy (1 week)
- Intestine lesion
 - Infection and destruction of enterocytes population the intestinal crypts > mucosal collapse, maldigestion and malabsorption diarrhea



Diagnosis



+

- Fecal enzyme immunoassays
 - Now facilitate rapid detection
 - Virus shedding is transient (between days 3-7 after infection)
- Direct examination
 - PCR
 - Hemagglutination
 - Agglutinated RBC of pig, cat or rhesus monkey by virus present in fecal extracts
 - Electron microscopy
- Virus culture and identification



Control and prevention

- Vaccination at 2-3 weeks interval
 - Starting at 6-8 weeks of ages
 - Continuing through 16-20 weeks of ages
- Modified live vaccine
- Disinfectant; 1% sodium hypochlorite





Porcine parvovirus

- Reproductive failure
- **SMEDI (stillbirth, mummification, embryonic death, infertility)**
- Infection of older swine
 - Causes only a mild or subclinical disease
- But the virus has also been associated more rarely with respiratory disease, vesicular disease, and systemic disease of neonates.

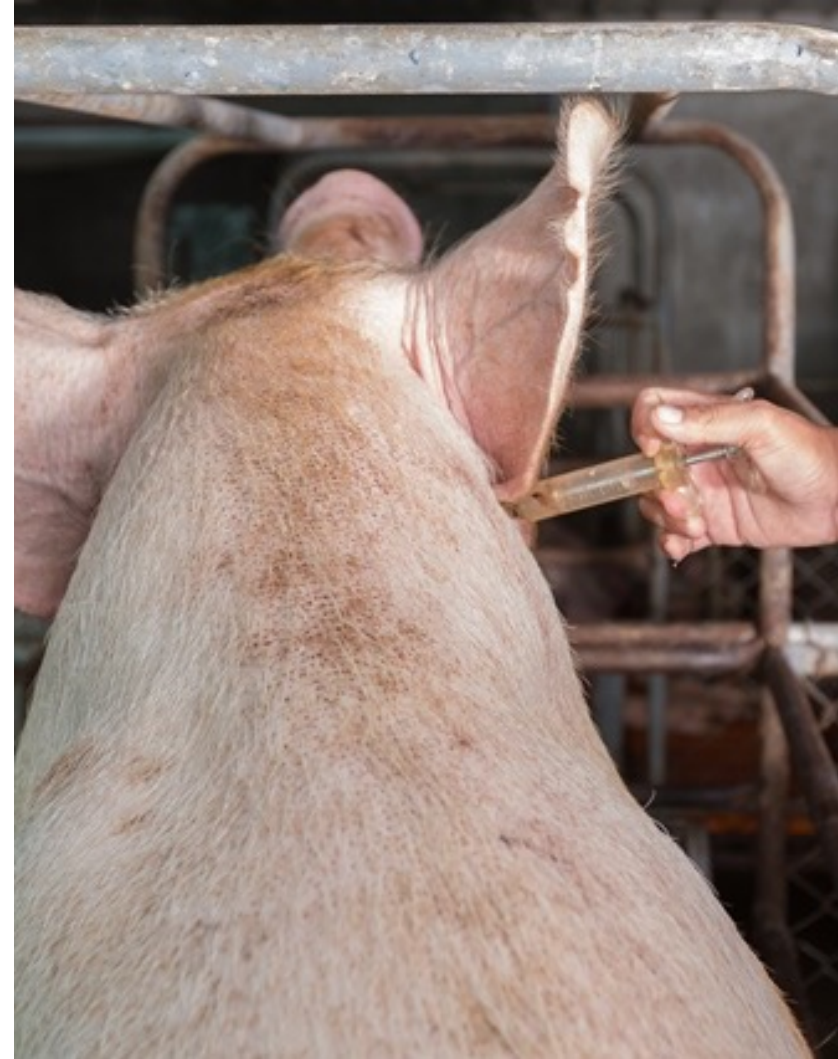


Diagnosis

- PCR assay: very sensitive, but the interpretation of results is important, as the assay may detect viral DNA even when the virus is not the primary cause of the disease
- Serologic tests are of limited value: virus is so widespread in swine, and vaccination may interfere
- Hemagglutination: agglutinated guinea pig RBC by virus contained in extracts of fetal tissues
- Diagnosis is difficult if infection occurs in the first few weeks of gestation
 - Fetuses are resorbed completely and no suspicion of the presence of the virus, and hence no specimens collected for laboratory diagnosis.

Control and prevention

- Keys to ensure good reproductive performance in breeding herds
 - Vaccination, to reduce transplacental infection, viral load and fetal mortality
 - Biosecurity
 - Hygiene, including cleaning and disinfection protocols of all buildings/rooms between batches of pigs

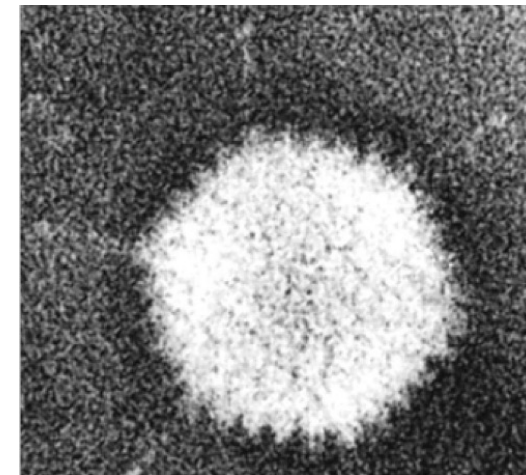
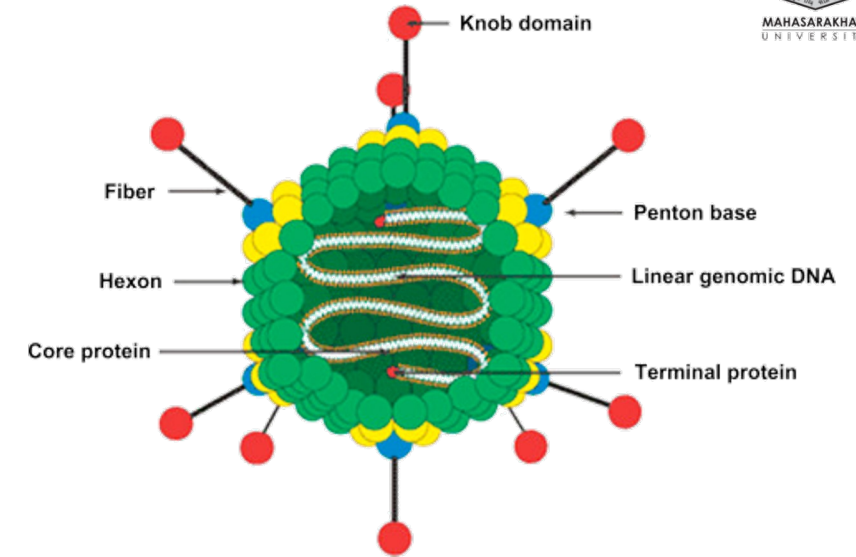


Family *Adenoviridae*

- **Taxonomic structure of the family**
- Family *Adenoviridae* comprise 4 Genus
 - **Genus *Mastadenovirus***
 - Infect only mammalian species
 - **Genus *Aviadenovirus***
 - Infect only birds
 - **Genus *Atadenovirus***
 - Infect a broad host range; including snakes, lizards, ducks, geese, chickens, possums, and ruminants
 - **Genus *Siadenovirus***
 - Frog adenovirus 1 and Turkey adenovirus 3

Virion properties

- Non-enveloped virus
- Icosahedral symmetry, 70-90 nm diameter, dsDNA; 36-44 kbp
- Capsid: 252 capsomers
 - 240 hexons เป็นผิวหน้าและขอบของชั้นสามเหลี่ยม
 - 12 pentons เป็น vertex แต่ละ penton จะมี penton fiber และมี knob ที่ปลาย
- Host range: single species/closely relate species
- Some viruses are oncogenic in laboratory animals
- Viruses agglutinate red blood cells
- Moderate stable in environment; ถูกทำลายได้ด้วย detergent ทั่วไป



Viral replication

• Baltimore classification

- Class I virus: double stranded DNA; dsDNA

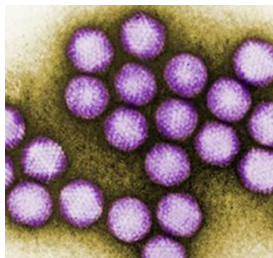
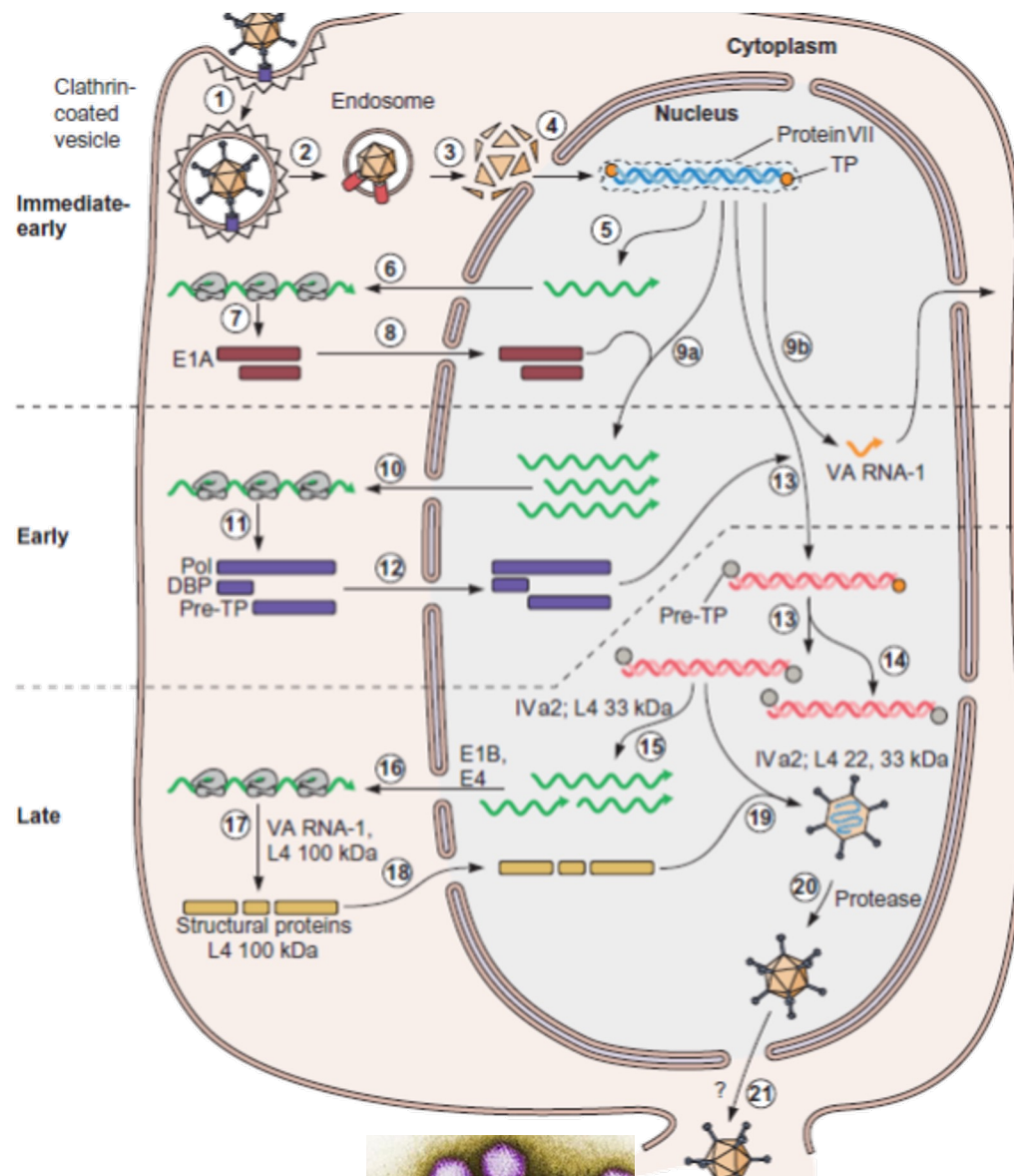
- Replication in nucleus; ทำให้เห็น inclusion bodies จำนวนมากใน NC

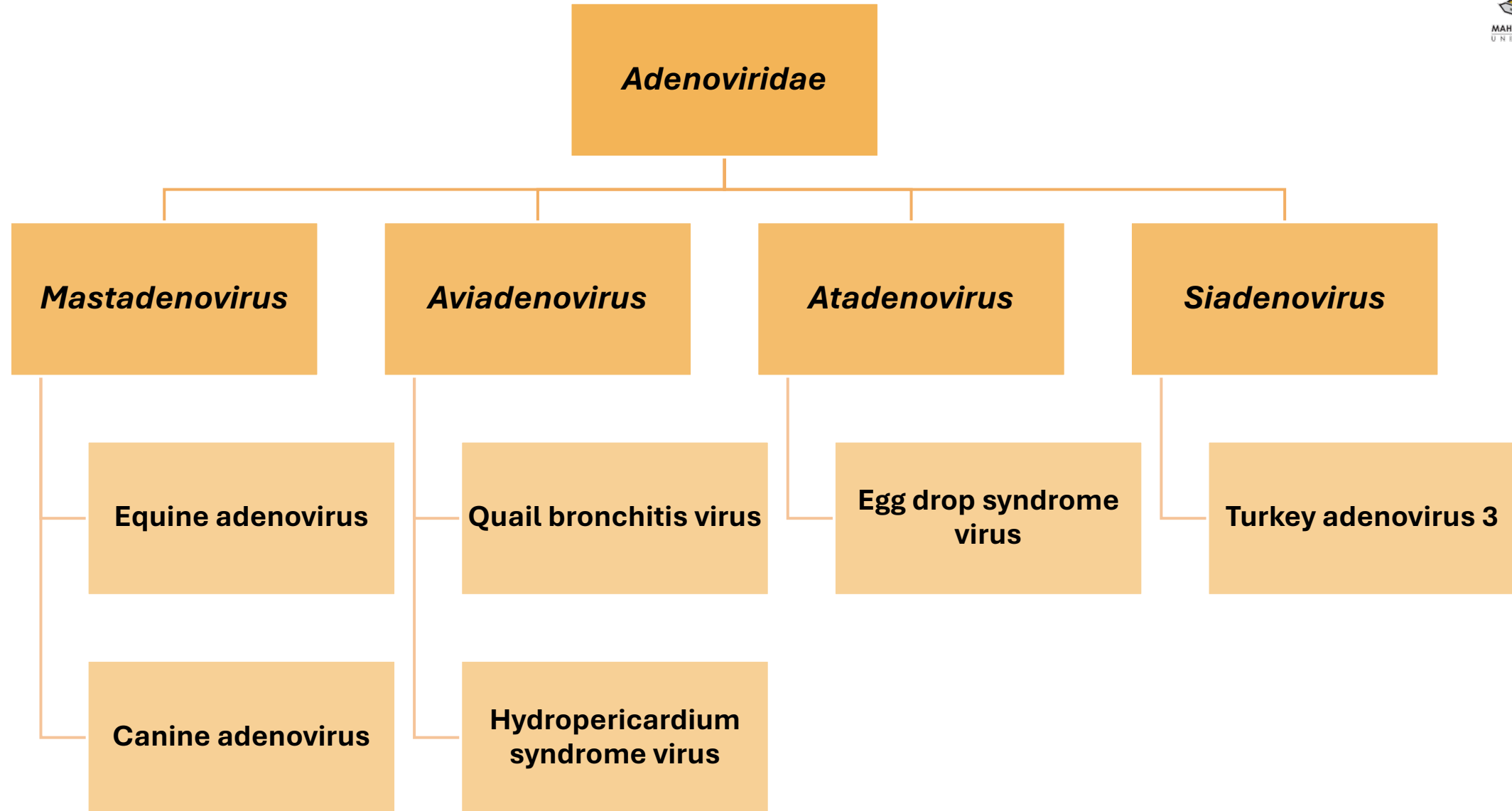
- ไวรัสจับกับ receptor ของเซลล์โฮสต์โดยใช้ penton fiber และผ่านเซลล์ด้วยขบวนการ endocytosis

- Pentons จะถูกปลดออกใน cytoplasm แล้วส่วนของ genome จะเข้าสู่ nucleus ของเซลล์

- Virion จะถูกประกอบขึ้นใน nucleus ซึ่งจะมีการสร้าง crystalline aggregates ทำให้ nuclei ผิดรูป ทำให้เห็น inclusion bodies ใน nucleus

- Virion จะถูกปล่อยออกจากเซลล์โดยขบวนการแตกของเซลล์ (cell lysis)





Genus *Mastadenovirus*



Canine adenovirus

- Canine adenovirus type 1
 - Infectious canine hepatitis; systemic disease
- Canine adenovirus type 2
 - Kennel cough syndrome; localized respiratory disease in dogs

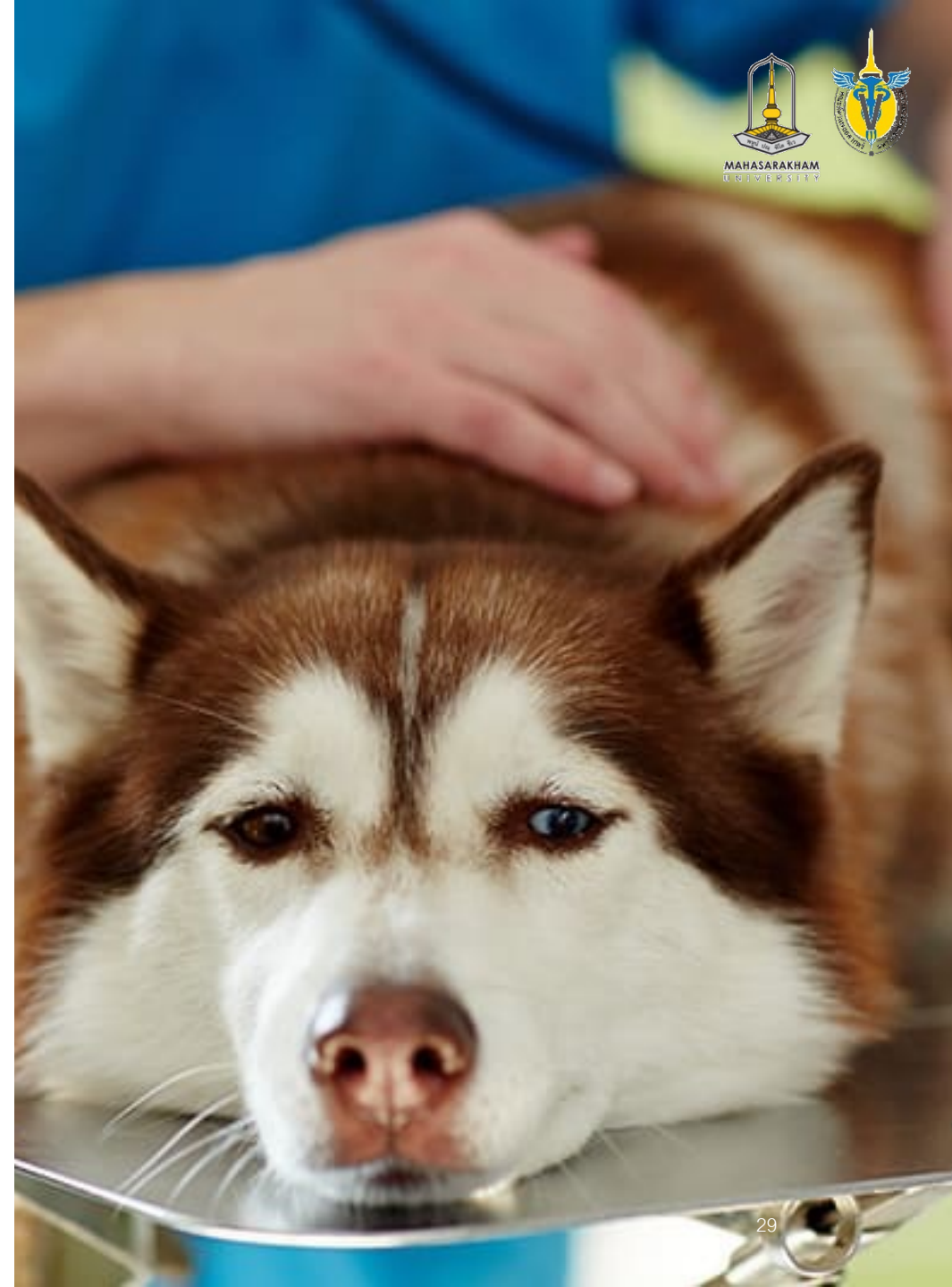


Equine adenovirus

- Equine adenovirus type 1
 - Isolated worldwide from upper respiratory secretions of foals and horses with and without disease
- Equine adenovirus type 2
 - Isolated from lymph nodes and feces of foals with upper respiratory disease and diarrhea

Canine adenovirus 1

- **Infectious canine hepatitis**
 - Systemic disease caused by canine adenovirus 1
 - Important pathogen of foxes, wolves, coyotes, skunks, and bears
 - In dogs, as well as causing acute hepatitis, the virus may cause respiratory or ocular disease
 - In contrast, canine adenovirus 2 infection is localized to the respiratory tract (as described in the subsequent section)
 - IP 4-9 days (acute infection)

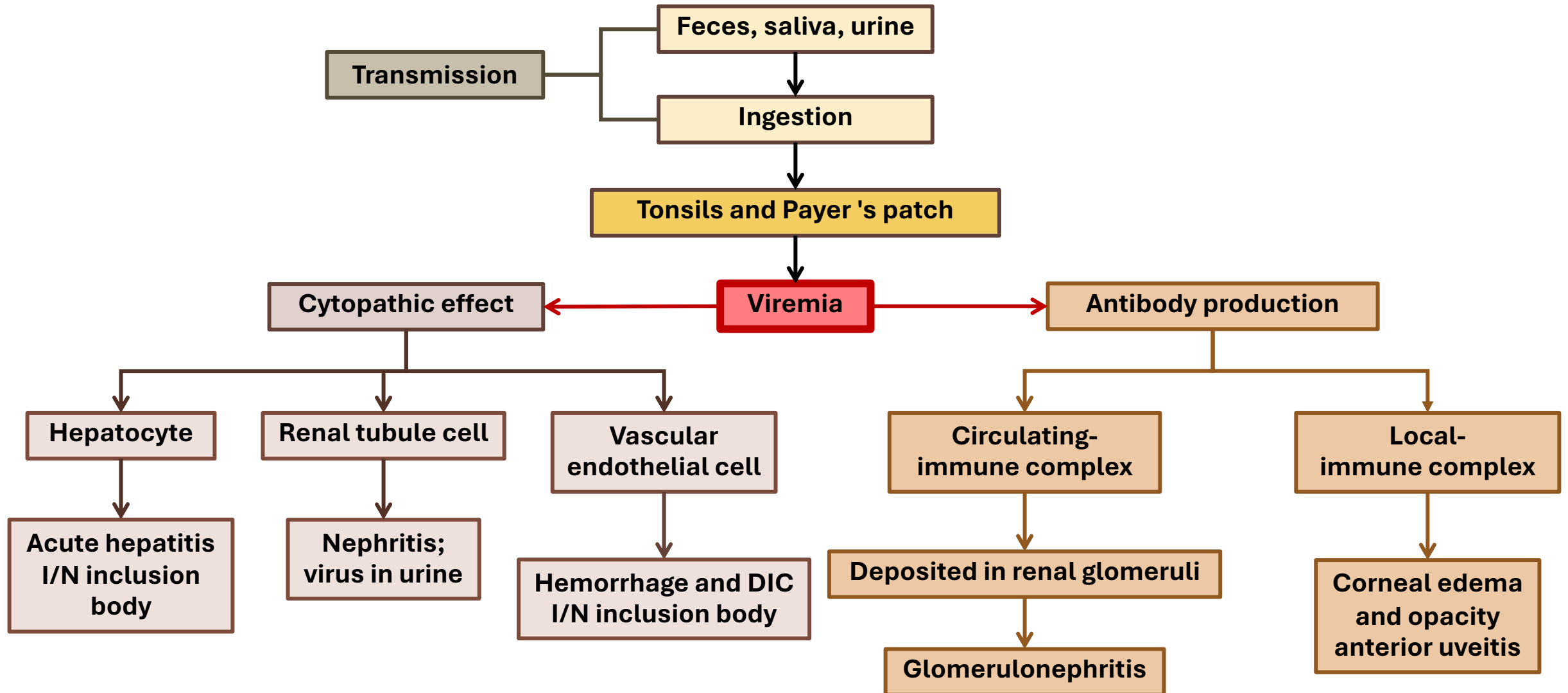


Clinical signs

- Fever, depression, anorexia, vomit, conjunctivitis, serous discharge from the eyes and nose
- Abdominal pain and petechiae of the oral mucosa
- Tachycardia, leukopenia, prolonged clotting time and disseminated intravascular coagulation (DIC)
- 7-10 days after acute signs disappear, about 25% of affected dogs develop a characteristic and diagnostically useful bilateral corneal opacity

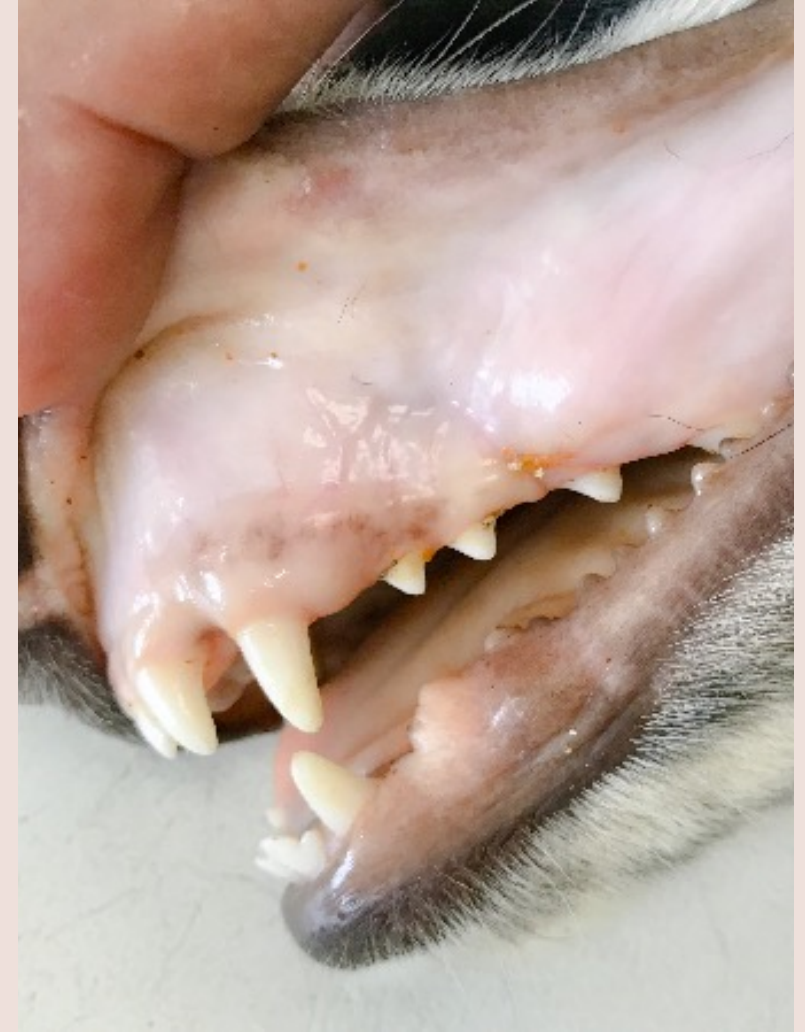


Pathogenesis



Systemic disease of ICH

- Infected in puppy < 6 months
 - **Per acute disease** – sudden death in 3 – 4 hours
 - **Acute disease** – fever, depression, loss of appetite, vomiting, bloody diarrhea, petechial hemorrhages of the gums, pale mucous membranes, and icterus (jaundice)
 - **Mild disease** – which may actually be a vaccine-modified disease that is, the result of partial immunity



Diagnosis

- Physical examination, hematological data, urinalysis, and postmortem findings
- Direct examination
 - PCR
 - FAT
- Antibody detection
 - ELISA
 - HI test
 - VN test
- Virus isolation

Canine adenovirus 2

- **Kennel cough syndrome**
 - Synonym: Infectious canine tracheobronchitis
 - Localized respiratory disease in dogs
 - Aerosol transmission > replication in both upper and lower respiratory tract
 - Clinical signs: respiratory tract disorders, tonsillitis, laryngitis, tracheitis, pneumonitis

Control and prevention

- The antigenic relationship between CAV-1 and 2 is sufficiently close for CAV-2 vaccine to be cross-protective
 - It has the advantage that it does not cause corneal edema
- The CAV-2 vaccine is frequently included in combination vaccines like Nobivac® Canine 1-DAPPv, which also protect against canine distemper, canine parvovirus, and canine parainfluenza virus
 - The primary vaccination series typically begins at 6-8 weeks of age, with booster shots given 2-4 weeks later
 - Maternal antibody interferes with active immunization until puppies are 9-12 weeks of age



Equine adenovirus



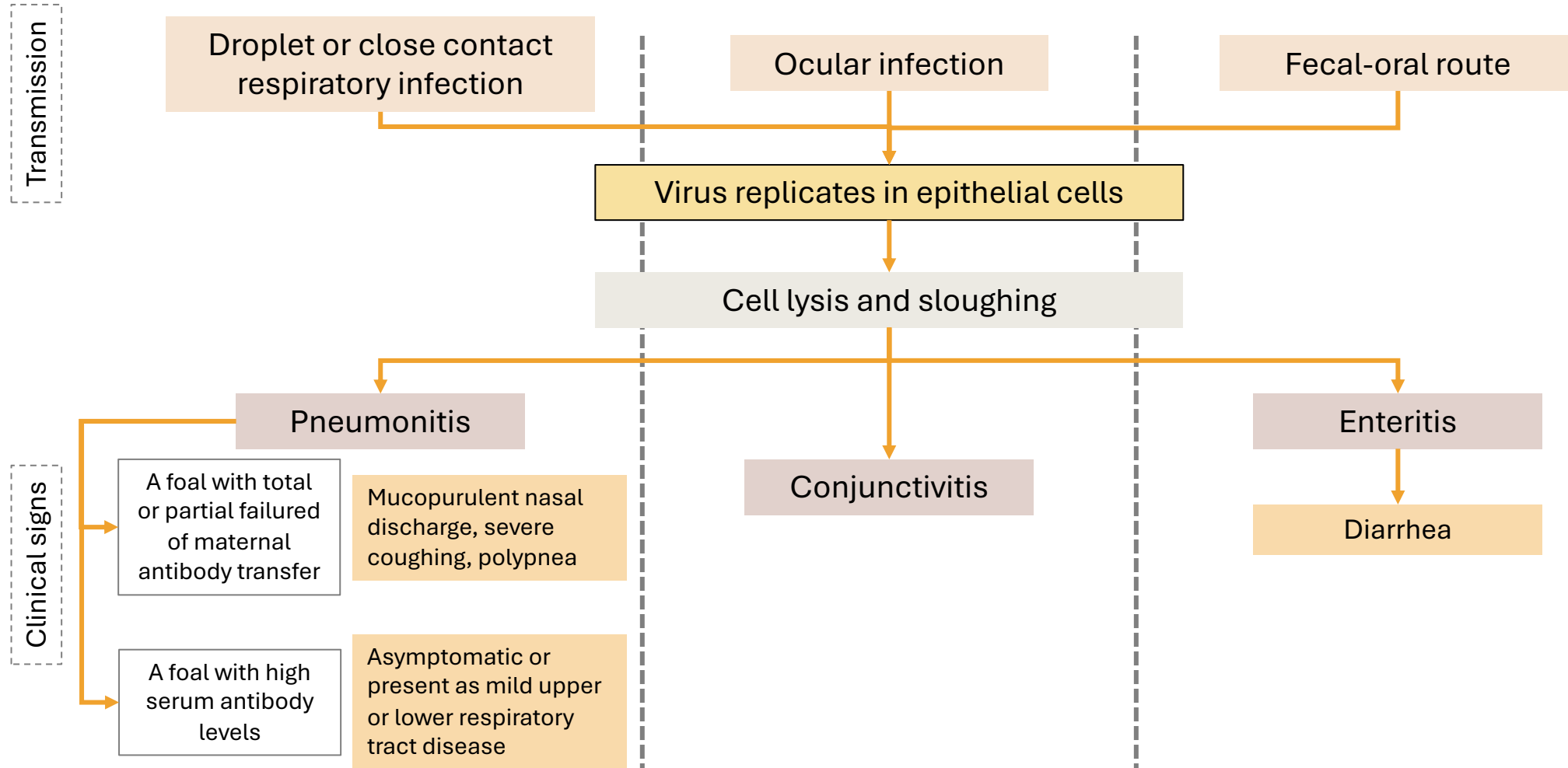
- Transmitted by oral, nasopharyngeal routes and ocular infection
- Most equine adenovirus infections are asymptomatic or present as mild upper or lower respiratory tract disease
 - In foals, EAdV-1 causes acute upper respiratory tract disease, follicular conjunctivitis, bronchopneumonia and infection of the gastrointestinal tract that is associated with production of soft feces
 - EAdV-2 has been isolated from diarrheic foal feces

Clinical signs and immunity

- Acute infection: fever, nasal discharge, coughing after exercise and enlarged submandibular lymph nodes
 - 2nd infection with bacteria or colostrum-deprived foal: mucopurulent nasal discharge, severe follicular conjunctivitis, transient anorexia and pyrexia and sustained polypnea
 - IgA in nasal secretions is responsible for resistance to reinfection
- Immune competent foals develop significant EAdV1 antibody, recover spontaneously from the disease (self-limited)



Pathogenesis



Recovery from disease caused by EAdV1 alone in immunologically competent horses occurs within a week or 10 days, but mixed infections with other viruses and bacteria may cause more severe and prolonged disease



Diagnosis

- Virus isolation from nasopharyngeal and conjunctival swabs during the acute phase of infection
- PCR – confirmed EAdV-infection
- Seroepidemiologic study and typing the EAdV based on
 - Serum neutralization assay
 - Hemagglutination-inhibition test
- Antibody detection: ELISA

Control and prevention

- Colostrum-fed
- Vaccination
 - Live EAdV-1 vaccine: intramuscular immunization (once)
 - Clinical disease did not develop, and the virus was not isolated
 - Inactivated EAdV-1 vaccine: a two-dose series, 2 weeks interval



Genus *Aviadenovirus*

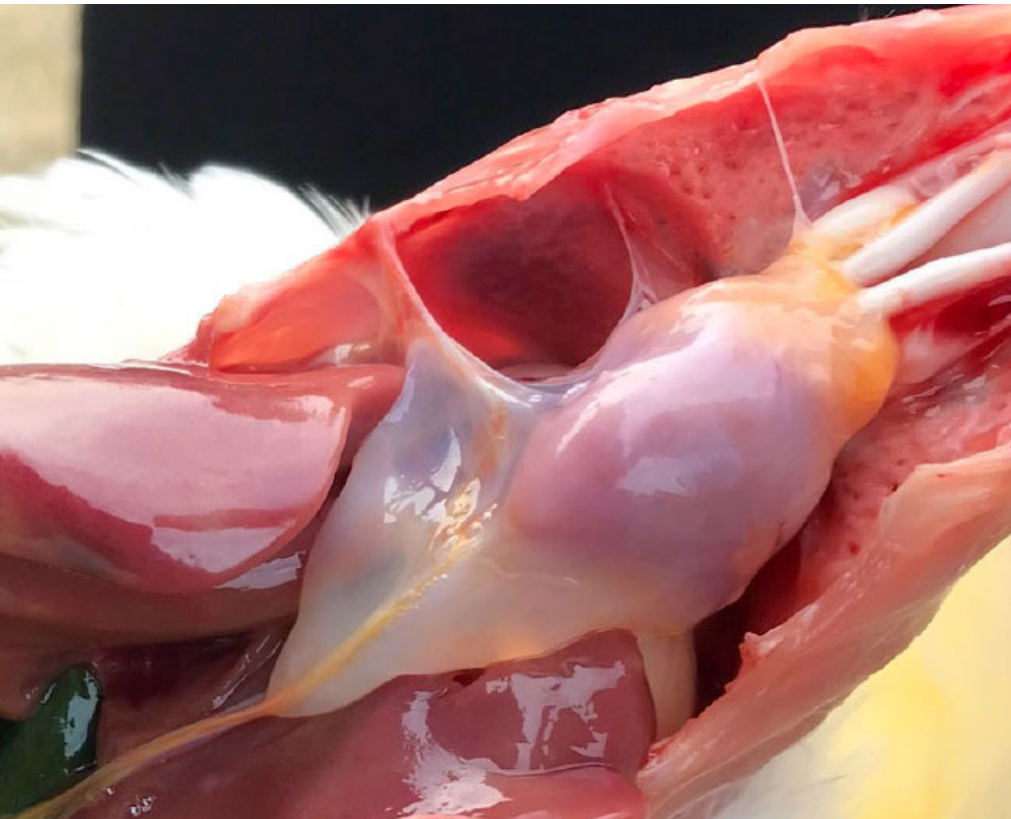
- Quail bronchitis virus
- Hydropericardium syndrome virus



Quail bronchitis virus

- Avian adenovirus 1: ก่อโรค Quail bronchitis ในนกกระทา
 - In young birds
 - open-mouth breathing, nasal discharge, coughing, sneezing, rales, lacrimation, and conjunctivitis
 - Mortality may be 100%
 - In older birds there is also diarrhea
 - Mortality 25% in birds aged more than 4 weeks
- Clinical sign: tracheitis, air-sacculitis and gaseous, mucoid enteritis
- Control is based on strict isolation, quarantine of introduced birds and regular decontamination of premises and equipment





Hydropericardium syndrome virus

- ก่อโรค Inclusion body hepatitis หรือ Hydropericardium syndrome หรือ Hepatitis hydropericardium
- 20-80% mortality, IP 3-7 days
- โดยมากพบโรคในไก่อายุน้อย ก่อโรคได้ทั้งในไก่, เป็ด, ห่าน, นกกระทา
- เชื้อสามารถถ่ายทอดผ่านทางไข่ได้ จึงมีผลต่อลูกไก่และก่อความเสียหายต่อการฟัก
- Pericardial effusion, pulmonary edema, hepatomegaly, and have enlarged kidneys
- A vaccine is available in some countries

Genus *Atadenovirus*

- **Egg Drop Syndrome**

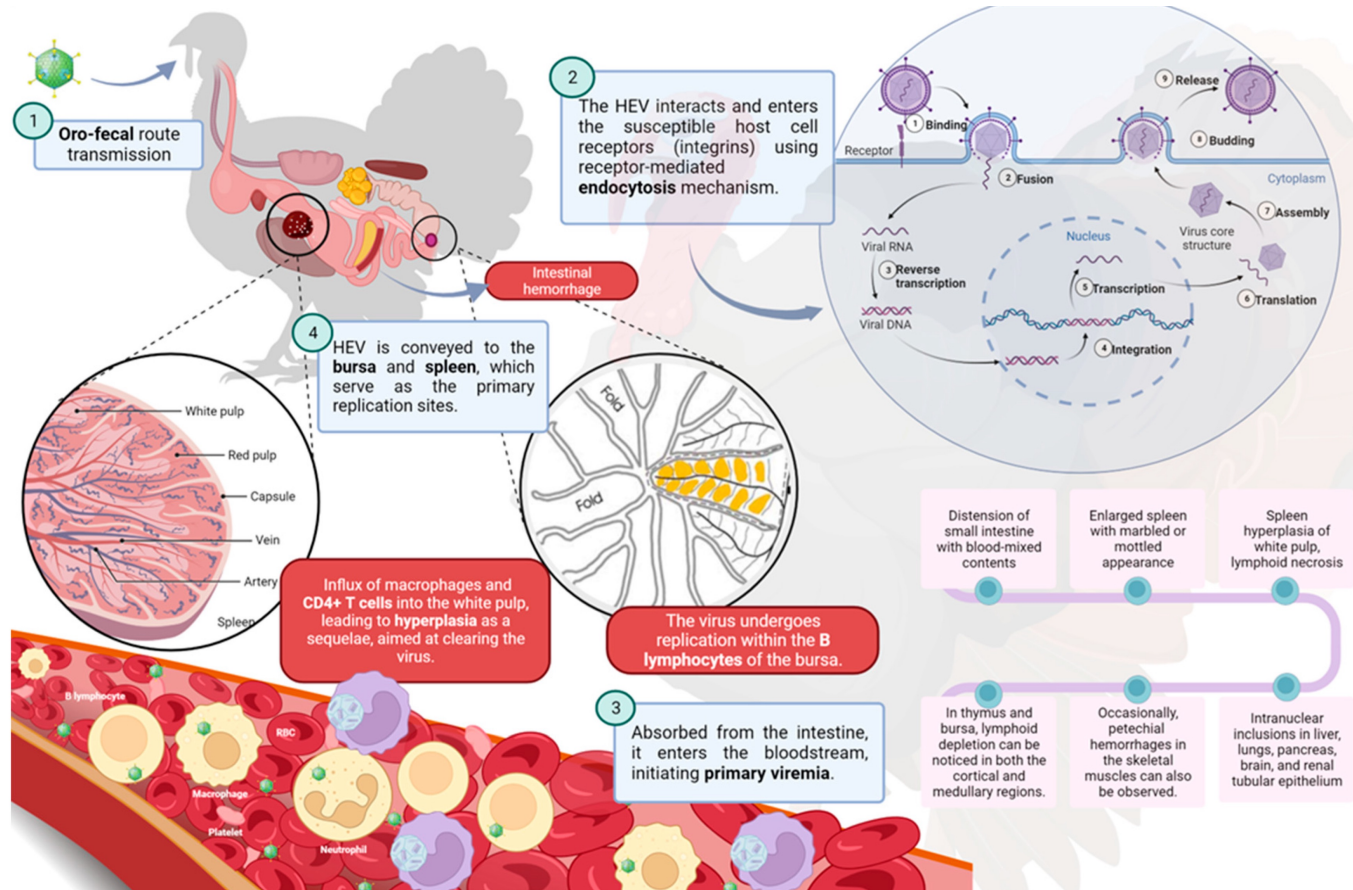
- Chickens are the major species affected by the disease
- Clinical signs of infection are loss of color in pigmented eggs and soft-shelled (ไข่เปลือกนิ่ม), thin-shelled (เปลือกบาง) and shell-less eggs (ไม่มีเปลือก)
- Egg production numbers decrease by a maximum of 40%
- Route of transmission: contaminate eggs, feces, contaminated fomites, needles used for vaccinations
- Major lesions in infected birds are seen in the pouch shell gland and oviduct, where epithelial cells become necrotic and contain intranuclear inclusion bodies
- Diagnosis: virus isolation or serology, HI test or VN test



Genus *Siadenovirus*

• Turkey Adenovirus 3

- การก่อโรค: Hemorrhagic Enteritis ในไก่วงวง, Marble Spleen Disease ในไก่ฟ้า
- Hemorrhagic enteritis is a common acute infection of turkeys older than 4 weeks
 - characterized by splenomegaly and intestinal hemorrhage
- Clinically, the disease is characterized by acute onset, depression, bloody droppings and death
- Infection causes both humoral- and cell-mediated immunosuppression
- Mortality 60%, although the usual mortality is 5-15%



Musa L, Rapi MC, Franciosini MP, Lupini C, Catelli E, Addis MF, Grilli G. Turkey Hemorrhagic Enteritis (THE): A Short Overview. *Pathogens*. 2024; 13(8):663. <https://doi.org/10.3390/pathogens13080663>

Diagnosis and control

- Diagnosis: Virus isolation, VN test, ELISA, AGID, IFA or PCR
- Control and prevention:
 - Strong biosecurity protocols such as deep physical cleansing, stringent sanitization of feeding equipment, and adequate ventilation
 - The disease in turkeys or pheasants is based on vaccination
 - วัคซีนเชื้อเป็น ให้โดยการผสมน้ำดื่ม เริ่มทำวัคซีนที่อายุ 4-5 สัปดาห์



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- Quinn, P.J., Markey, B.K., Leonard, F.C., FitzPatrick, E.S., Fanning, S. and Hartigan, P.J. 2011. Veterinary Microbiology and Microbial Disease. 2nd edition. Wiley-Blackwell, West Sussex, UK. 912p.

